# PAPER_929: Neutron Star Phonon Spindown Correction

**Author:** Daniel T. Murphy – Star Magic / UQFF Framework **Date:** 2026-04-11 **Session:** 211 **Source:** SCm phonon gap implementation (ns_phonon_gw190425_wstp.py) **Calculator:** NSPhononSpindownCorrectionCalc (CP4 #513) **CVW:** v2.0.0 compliant

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## Abstract

Derives the phonon-corrected neutron star spindown rate: Omega_dot_NS^phonon = Omega_dot_NS * (1 + Phi*S_26*[SSq]/N), where the phonon modulation Phi couples to the 26-layer polylog sum S_26 to create an additional angular momentum loss channel. The correction factor Phi*S_26*[SSq]/N represents phonon-mediated vacuum dissipation that enhances the standard magnetic dipole braking torque. For canonical pulsars (Omega_dot ~ -4.2e-15 rad/s^2), the phonon correction is enormous at resonance (Phi ~ 10^20), implying that phonon-dominated spindown would deplete angular momentum far faster than magnetic dipole radiation alone. This constrains the effective phonon coupling in real NS environments.

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## 1. Core Equations

### Section A: Lagrangian

```
Omega_dot_NS^phonon = Omega_dot_NS * (1 + Phi * S_26 * [SSq] / N)
Phi = Phi_0 * exp[-(omega - omega_SCm)^2 / (2*Gamma^2)]
S_26 = sum_{k=1}^{26} exp(-[SSq]*k/26)
N = 26  (number of UQFF layers)
```

### Section B: VDS/DVP/BH Number Systems

```
Braking index: n_phonon = n_dipole * (1 + phonon_correction)
Characteristic age: tau_phonon = P / (2 * |Omega_dot_phonon|)
Magnetic field: B_phonon = B_dipole / sqrt(1 + correction)
```

### Section SM: SM Anchors

```
Omega_dot_NS = -4.2e-15 rad/s^2  (canonical pulsar)
[SSq] = 0.57
N = 26 layers
omega_SCm = 2*pi * 1.25e12 rad/s
```

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## 2. Parameters

| Parameter | Default | Description |
|---|---|---|
| Omega_dot_NS | -4.2e-15 | Base spindown (rad/s^2) |
| Phi_0 | 10^20 | Peak phonon amplitude |
| omega | omega_SCm | Phonon frequency |
| N_layers | 26 | UQFF layers |

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## 3. Key Results

| Omega_dot_NS | Correction | Enhancement |
|---|---|---|
| -1e-16 | Phi*S_26*[SSq]/26 | »1x at resonance |
| -4.2e-15 | same | »1x at resonance |
| -1e-13 | same | »1x at resonance |

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## 4. Physical Interpretation

The phonon spindown correction reveals that at full 1.25 THz resonance, the correction factor overwhelms the base magnetic dipole torque. This implies that real NS environments must be significantly off-resonance (omega != omega_SCm) or that the effective Phi is greatly reduced by NS interior conditions (superconducting proton fluid, superfluid neutron component). The correction provides a new mechanism for anomalous braking indices (n != 3) observed in young pulsars, and constrains the phonon coupling strength from timing measurements.

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## 5. References


- PAPER_914: Phonon-corrected NS spindown (Session 210b)
- PAPER_915: Magnetar spin-down phonon timescale
- ns_phonon_gw190425_wstp.py: 5-class standalone module
- WSTP expression #31: Omega_dot_NS^phonon